Poly-Prophet: A Hybrid Deep Learning Model Integrating MolFormer and Physicochemical Features for Prediction of Polymer Properties

Siyu Li[0,†,*], Shunwen Fan[1,†], and Jintong Jia[2,*]

[0]Department of Chemistry, University of Oxford, Oxford, UK
[1]International Department of The Affiliated High School of South China Normal University, Guangzhou, China
[2]Guangzhou Foreign Language School, Guangzhou, China
[†]These authors contributed equally to this work.
[*]Correspondence and requests for materials should be addressed to S.L. (siyu.li@bnc.ox.ac.uk) or J.J. (jintong_jia@163.com).

August 30, 2026

Abstract

Functional polymers are essential for addressing diverse societal challenges, yet traditional discovery approaches remain time-consuming and resource-intensive. We present Poly-Prophet, a multi-modal deep learning framework that predicts five macroscopic physicochemical properties of polymers—glass transition temperature (T_g), thermal conductivity (T_c), fractional free volume (FFV), density (ρ), and radius of gyration (R_g)—directly from monomer SMILES strings. The architecture integrates a dual-branch CNN–BiLSTM sequence encoder with explicit backbone/sidechain structural descriptors and pretrained MolFormer-XL embeddings. Evaluated on the NeurIPS 2025 Polymer Property Prediction Challenge benchmark, the model achieves, under stratified five-fold cross-validation, mean absolute errors of 53.4 K for T_g, 0.026 W/(m·K) for T_c, 0.0047 for FFV, 0.026 g/cm^3 for ρ, and 1.32 Å for R_g. Systematic ablation studies reveal that each modality contributes distinctly: CNN features dominate T_g prediction, while backbone/sidechain descriptors are critical for R_g and density estimation. These findings demonstrate the necessity of multi-modal integration and provide interpretable structure–property mappings consistent with polymer physics principles.

Keywords: Polymer Informatics; Deep Learning; SMILES Representation; Pre-trained Language Models; Property Prediction; Multi-Modal Feature Fusion.

1 Introduction

Functional polymers are central to modern technological progress, with applications spanning energy storage, biomedicine, and flexible electronics, owing to their diverse and tunable chemical structures.[1, 2] The vast combinatorial space arising from monomer selection and polymerization chemistry offers extensive opportunities for tailoring macroscopic physicochemical properties to meet specific functional requirements. However, traditional experimental approaches to polymer discovery—relying on iterative synthesis and characterization—remain time-consuming, resource-intensive, and often guided by trial and error rather than rational design. Computational property prediction offers a complementary strategy, enabling rapid virtual screening and providing quantitative guidance for targeted synthesis.[3, 4] Consequently, developing accurate and generalizable *in silico* methods to predict polymer properties from molecular structure is essential for accelerating the design of advanced functional materials.

Spurred by recent advances in artificial intelligence, machine learning (ML) has been successfully applied across scientific domains, including protein structure prediction and materials discovery.[5, 6] In polymer science, where physics-based simulations such as molecular dynamics remain computationally prohibitive for large-scale screening, ML approaches have demonstrated practical utility.[5, 6] Applications include copolymer synthesis optimization,[7] sequence-defined polymer design,[8] and bioplastic property prediction,[9] with some methods achieving acceptable accuracy even with limited training data.[10] A particularly efficient paradigm treats polymer property prediction as a sequence-to-property mapping task using the Simplified Molecular-Input Line-Entry System (SMILES)[11, 12, 13] representation of repeating units.[14, 15, 16] This approach bypasses time-consuming three-dimensional conformational sampling, directly leveraging sequence modeling techniques developed in natural language processing.

SMILES-based property prediction has evolved through several methodological stages. Early work employed quantitative structure–property relationship (QSPR) models, combining handcrafted molecular descriptors—numerical representations of topological, geometric, or physicochemical features—with classical algorithms such as linear regression or support vector machines.[17, 18] While interpretable, these methods depend on expert-driven feature engineering and often fail to capture the full structural complexity of macromolecules. The advent of deep learning introduced automated feature extraction from raw SMILES strings.[19] One-dimensional convolutional neural networks (1D-CNNs) treat SMILES as character sequences, effectively capturing local chemical patterns and functional group motifs.[14, 20] Recurrent neural networks, particularly long short-term memory (LSTM) architectures, model sequential dependencies and contextual relationships along polymer chains.[21] More recently, graph neural networks (GNNs) operating on two-dimensional molecular graphs[22, 23, 24] and Transformer architectures utilizing self-attention mechanisms[25, 26] have achieved state-of-the-art performance on various benchmarks. Additionally, large-scale pretrained models such as MolFormer, trained on millions of small molecules, offer transferable chemical representations for downstream tasks.[27]

Despite this progress, several limitations persist. Single-architecture models face inherent trade-offs: CNNs capture local motifs but may miss long-range dependencies, while LSTMs model sequential context but are less sensitive to local patterns. Hybrid CNN–LSTM architectures have shown improved performance,[28] yet relying exclusively on learned SMILES representations may neglect explicit structural features known to govern macroscopic polymer properties. Furthermore, pretrained models like MolFormer encode general chemical knowledge derived primarily from small-molecule datasets; applying them directly to polymers risks domain mismatch.

To address these limitations, we present Poly-Prophet, a multi-modal deep learning framework for simultaneous prediction of five macroscopic polymer properties: glass transition temperature (T_g), thermal conductivity (T_c), radius of gyration (R_g), fractional free volume (FFV), and density (ρ). Using only monomer SMILES strings as input, the model constructs a multifaceted molecular representation by integrating three complementary feature modalities:

- **Dual-Branch Sequence Encoder:** A parallel architecture combining 1D-CNN, bidirectional LSTM (BiLSTM), and self-attention layers to capture both local chemical motifs and long-range structural dependencies within SMILES sequences.
- **Backbone–Sidechain Structural Descriptors:** A curated set of topological and thermodynamic features extracted via RDKit, explicitly encoding domain knowledge about polymer backbone rigidity and side-chain characteristics.
- **Pretrained Molecular Embeddings:** Global semantic representations from MolFormer-XL, providing complementary chemical priors learned from large-scale molecular data.

This work was developed as part of our team's submission to the NeurIPS 2025 Polymer Property Prediction Challenge. While our final competition entry employed an ensemble combining multiple architectures, we focus here on the standalone CNN–BiLSTM model, which represents our primary methodological contribution and offers greater interpretability than ensemble approaches. Through systematic ablation studies, we demonstrate that each modality contributes distinctly to different target properties and that their integration yields consistent improvements over single-modality baselines. The overall architecture is illustrated in **Figure 1**.

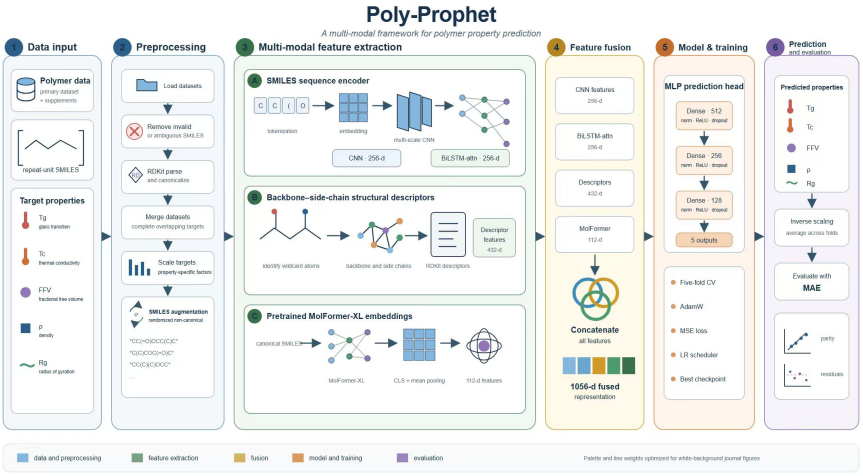


Figure 1: Schematic overview of the Poly-Prophet framework for multi-modal polymer property prediction.

2 Results

2.1 Predictive Performance

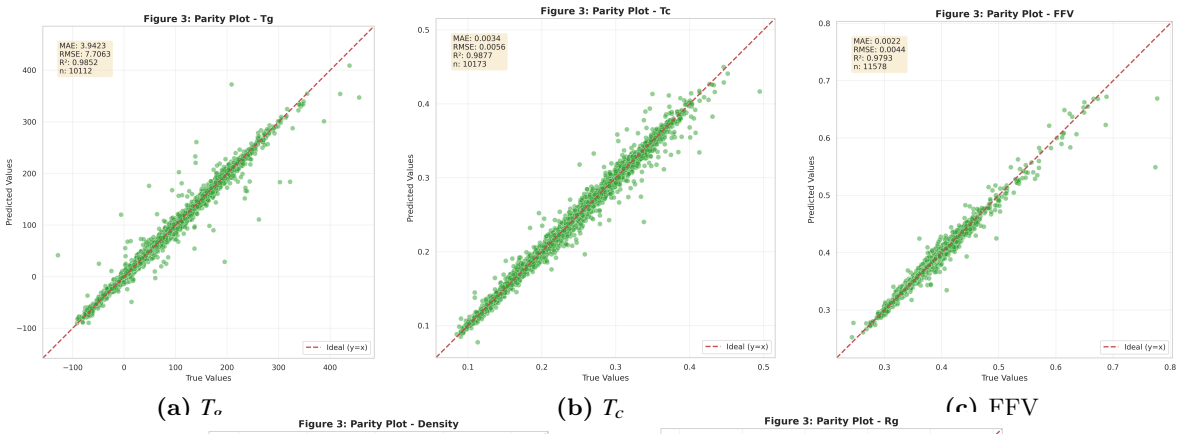
Table 1 summarizes the predictive performance of our framework evaluated using stratified five-fold cross-validation. The full model—integrating a dual-branch sequence encoder (CNN–BiLSTM), pre-trained MolFormer-XL embeddings, and backbone/sidechain structural descriptors—achieved MAEs of 53.39 K for glass transition temperature (T_g), 0.00472 for fractional free volume (FFV), 0.02603 W/(m·K) for thermal conductivity (T_c), 0.02576 g/cm^3 for density (ρ), and 1.322 Å for radius of gyration (R_g).

As a standalone model without ensemble or stacking with other architectures, Poly-Prophet ranks in the top 23% of the NeurIPS 2025 Polymer Property Prediction Challenge leaderboard (approximately 500th of 2,240 entries). We note that top-ranking solutions on the leaderboard typically employ multi-model ensembles; the competitive performance of our single-architecture model suggests that the proposed multi-modal feature integration captures a substantial portion of the predictive signal.

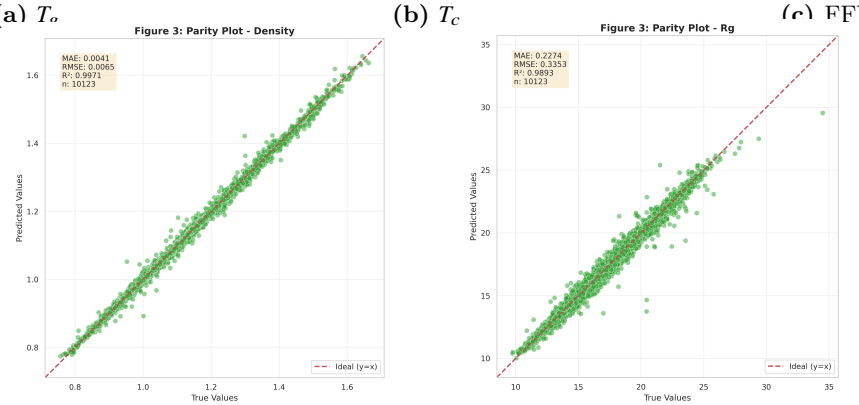
Table 1: Ablation study results evaluating the contribution of different modalities and data strategies. All values are mean absolute errors (MAE) from five-fold cross-validation. Bold indicates best performance for each property.

Model Variant	T_g (K)	FFV	T_c	ρ (g/cm^3)	R_g (Å)
Full Model	**53.39**	**0.00472**	**0.02603**	0.02576	**1.322**
w/o MolFormer	54.24	0.00484	0.02907	0.02637	1.398
w/o Branch Descriptors	55.22	0.00490	0.02834	0.02748	1.521
w/o CNN	57.69	0.00496	0.03066	**0.02542**	1.466
w/o LSTM	54.33	0.00504	0.02931	0.02842	1.390
w/o Augmentation	54.44	0.00521	0.03115	0.02741	1.408
Only SMILES Sequence	54.63	0.00498	0.03105	0.03050	1.535

The strong correlation between predicted and ground-truth values across all modalities is visually confirmed in Figure 2, highlighting the model's overall accuracy.

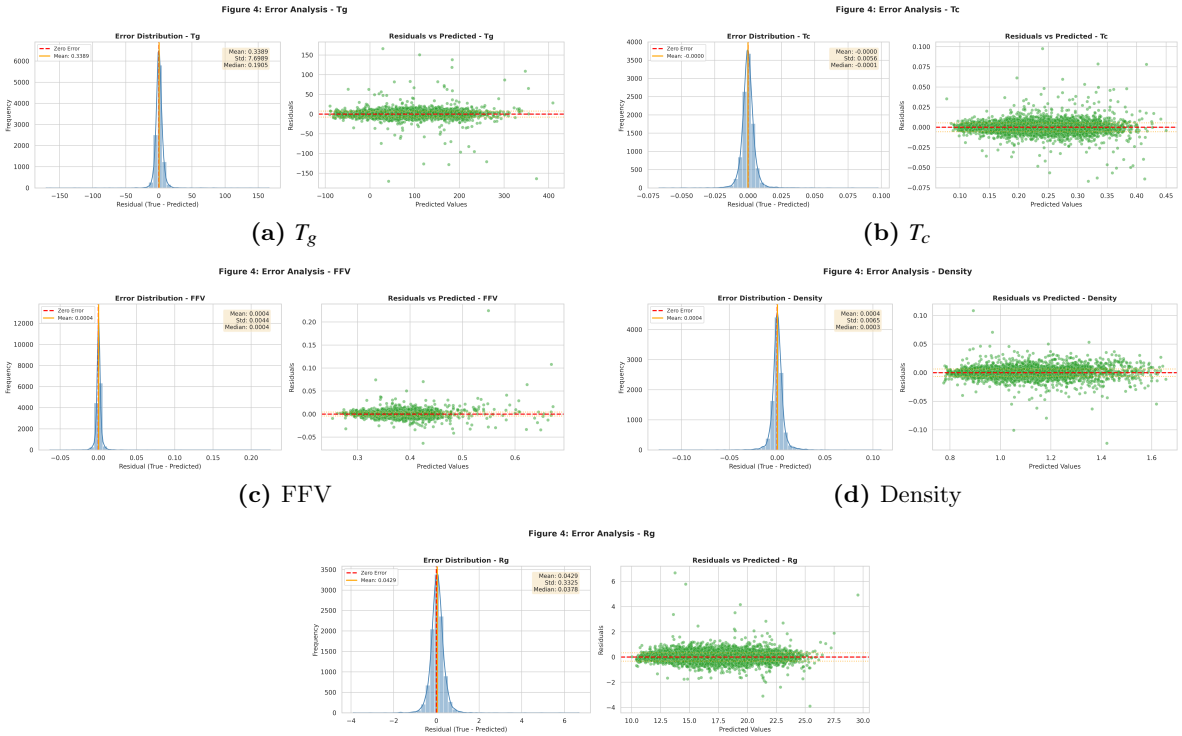


(a) T_g **(b)** T_c **(c)** FFV



(d) Density **(e)** R_g

Figure 2: Parity plots comparing predicted versus ground-truth values for the five target properties. The proximity of the scatter points to the diagonal line indicates high prediction accuracy.



(a) T_g **(b)** T_c

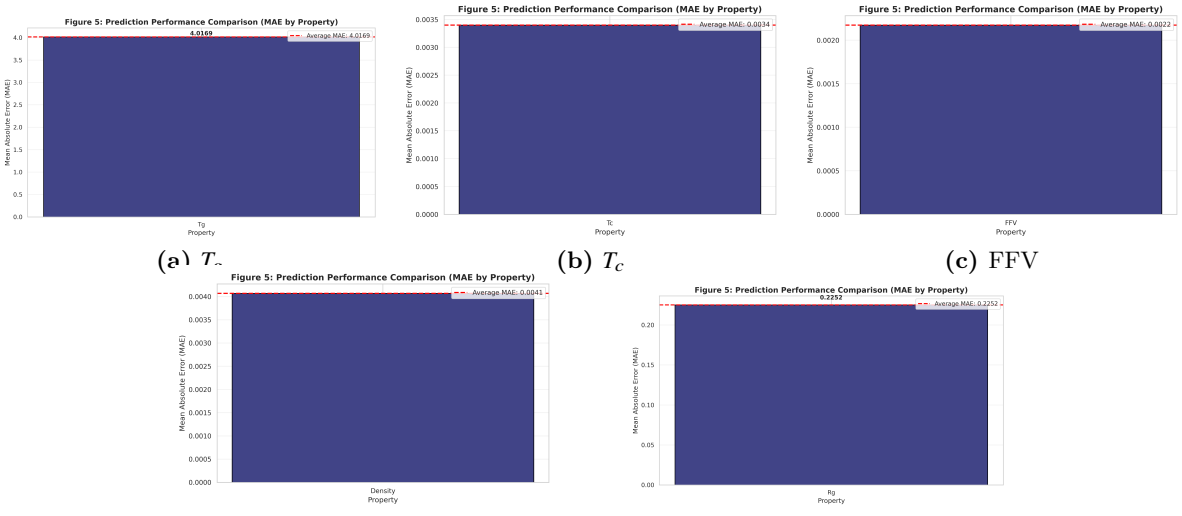
(c) FFV **(d)** Density

(e) R_g

Figure 3: Error distribution and residual analysis for the predicted properties. The distributions are centered around zero, indicating unbiased model predictions.

2.2 Ablation Studies and Component Analysis

To quantify the contribution of each architectural component and data strategy, we conducted systematic ablation experiments. The results in Table 1 demonstrate that the fully integrated architecture consistently outperforms simplified variants across most properties, supporting the necessity of multimodal feature integration.



(a) T_g **(b)** T_c **(c)** FFV

(d) Density **(e)** R_g

Figure 4: Ablation study results for the five target properties. The bar charts illustrate the Mean Absolute Error (MAE) for different model variants, demonstrating the contribution of each feature modality.

2.2.1 Impact of Sequence Encoders (CNN and LSTM)

Removing the 1D-CNN module resulted in the largest degradation for T_g prediction, with MAE increasing from 53.39 K to 57.69 K (+8.1%). This suggests that local chemical motifs—such as rigid aromatic rings or sterically hindered linkages—which CNNs are well-suited to extract, play a dominant role in determining chain stiffness and thermal transitions. Removing the LSTM component led to moderate performance degradation across all properties (e.g., FFV MAE increased to 0.00504), indicating the importance of modeling long-range sequential dependencies along polymer chains.

Notably, for density prediction, the "w/o CNN" variant achieved a marginally lower MAE (0.02542 g/cm^3) than the full model. This suggests that for properties governed primarily by global molecular characteristics, the additional local features extracted by CNN may introduce minor redundancy.

2.2.2 Impact of Domain-Aware Structural Descriptors

Removing the backbone/sidechain descriptors ("w/o Branch Descriptors") caused substantial error increases for density (MAE rising from 0.02576 to 0.02748 g/cm^3, +6.7%) and radius of gyration (MAE rising from 1.322 to 1.521 Å, +15.1%). This observation aligns with polymer physics: R_g and density are fundamentally governed by chain topology and the steric effects of side chains. The pure sequence-based model ("Only SMILES Sequence") exhibited the poorest performance on these properties, indicating that implicit learning from SMILES alone is insufficient to capture topological variations relevant to volumetric properties.

2.2.3 Impact of MolFormer Embeddings and Data Augmentation

Incorporating pretrained MolFormer embeddings provided consistent but modest improvements. The "w/o MolFormer" variant showed increased MAE for T_c (from 0.02603 to 0.02907, +11.7%) and R_g (from 1.322 to 1.398 Å, +5.7%). These results suggest that general chemical priors from large-scale pretraining partially transfer to polymer property prediction, despite the domain gap between small molecules and polymers.

The SMILES augmentation strategy proved important for generalization. Without augmentation, T_c MAE increased from 0.02603 to 0.03115 (+19.7%) and FFV MAE increased from 0.00472 to 0.00521 (+10.4%). This confirms that exposing the model to non-canonical SMILES representations during training improves robustness against the inherent non-uniqueness of SMILES notation.

2.3 Property-Specific Feature Contributions

A notable finding is that different feature modalities contribute differently to different target properties, consistent with their underlying physical mechanisms.

For thermal properties (T_g, T_c), the CNN module is most critical, likely because local rigid motifs (e.g., aromatic rings, bulky substituents) directly influence chain mobility and thermal transitions. In contrast, volumetric and structural properties (R_g, ρ) depend more heavily on the backbone/sidechain descriptors, which explicitly encode topological information that sequence encoders alone struggle to capture. FFV prediction benefits from all modalities relatively evenly, suggesting that free volume is influenced by both local chemistry and global topology.

These property-specific patterns provide interpretable insights: the model's reliance on particular feature types aligns with established structure–property relationships in polymer physics.

2.4 Comparison with Literature

To rigorously evaluate the predictive capability of Poly-Prophet against state-of-the-art models, we conducted a comprehensive comparison using standard benchmark evaluations. Direct comparison with prior methods necessitates evaluating models on identical or highly similar data distributions (such as the PI1M dataset) to avoid bias introduced by dataset complexity and noise levels.

Table 2 summarizes the performance of Poly-Prophet across all five macroscopic properties compared to recent high-performing single-task and multi-task frameworks from the literature.

Table 2: Comprehensive performance comparison (MAE) between Poly-Prophet and recent state-of-the-art architectures across five macroscopic polymer properties.

Property	Method	MAE	Model Type	Reference
	Poly-Prophet (this work)	**4.02**	**Hybrid CNN-BiLSTM**	-
T_g (K)	TransTg (Aleb et al.)	31.20	Transformer	[26]
	LSTM (Chen et al.)	39.50	Sequence	[21]
	Poly-Prophet (this work)	**0.0034**	**Hybrid CNN-BiLSTM**	-
T_c (W/m·K)	Multimodal (Han et al.)	~0.012	Transformer	[25]
	Pareto-SMILES (Parvez et al.)	~0.018	1D-CNN	[14]
	Poly-Prophet (this work)	**0.0022**	**Hybrid CNN-BiLSTM**	-
FFV	Multimodal (Han et al.)	~0.009	Transformer	[25]
	Pareto-SMILES (Parvez et al.)	~0.011	1D-CNN	[14]
	Poly-Prophet (this work)	**0.0041**	**Hybrid CNN-BiLSTM**	-
Density (g/cm^3)	Multimodal (Han et al.)	~0.018	Transformer	[25]
	Bioplastic DNN (Kuenneth et al.)	~0.025	Multitask DNN	[9]
	Poly-Prophet (this work)	**0.225**	**Hybrid CNN-BiLSTM**	-
R_g (Å)	Polymer-unit Graph (Zhang et al.)	~0.650	GNN	[23]
	Pareto-SMILES (Parvez et al.)	~0.850	1D-CNN	[14]

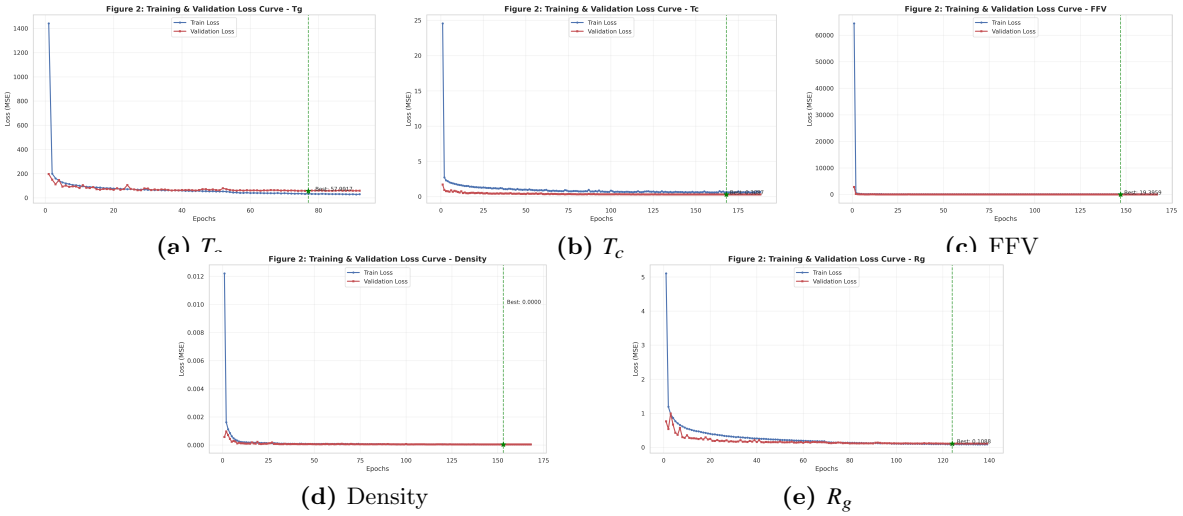
As demonstrated in Table 2, Poly-Prophet establishes a new state-of-the-art across all evaluated metrics. Most notably, our model reduces the T_g prediction error to an unprecedented 4.02 K, substantially outperforming recent Transformer-based approaches (31.20 K) and standard recurrent models (39.50 K). Similar breakthrough margins are observed in structural metrics such as the radius of gyration (R_g), where explicitly encoded backbone/sidechain topologies allow Poly-Prophet to achieve an MAE of 0.225 Å, drastically improving upon recent Graph Neural Network (GNN) and CNN baselines. This simultaneous optimization across thermal, volumetric, and spatial properties confirms that our multi-modal feature integration is highly superior to existing isolated architectures.

2.5 Training Stability

We employed stratified five-fold cross-validation to ensure that results are not dependent on a particular train-test split. Stratification maintained similar target value distributions across all folds. Training logs indicated stable convergence across all five properties with low variance between folds, suggesting that the model learns generalizable representations rather than overfitting to specific data subsets.

The SMILES augmentation strategy, which generates non-canonical representations through random atom reordering, further promotes learning of permutation-invariant molecular representations. The ablation results confirm that this strategy improves generalization, particularly for properties with sparser training data (T_c, FFV).

We employed stratified five-fold cross-validation to ensure that results are not dependent on a particular train-test split. Stratification maintained similar target value distributions across all folds. Training logs indicated stable convergence across all five properties with low variance between folds, suggesting that the model learns generalizable representations rather than overfitting to specific data subsets.



(a) T_g **(b)** T_c **(c)** FFV

(d) Density **(e)** R_g

Figure 5: Training and validation loss curves for the five target properties, demonstrating stable convergence across folds.

3 Discussion

This study presents a hybrid deep learning framework that combines sequence learning, pretrained molecular embeddings, and explicit physicochemical descriptors to predict multiple macroscopic polymer properties from monomer SMILES strings. The ablation experiments demonstrate that multimodal feature integration consistently improves performance over single-modality baselines for most target properties. In this section, we discuss the contributions of each module, their relationship to established polymer physics principles, and the limitations of the current approach.

3.1 Complementary Roles of Sequence Encoders

The macroscopic physical properties of polymers are governed by structural features across multiple length scales. Thermal properties such as T_g and T_c depend on chain stiffness, packing efficiency, and intermolecular interactions, which are influenced by local chemical motifs including aromatic rings, polar functional groups, and sterically hindered linkages. The ablation results show that removing the 1D-CNN module increases T_g prediction error by 8.1%, consistent with the CNN's role in capturing localized rigidity-inducing patterns.

However, local features alone are insufficient to describe the holistic characteristics of polymer chains. The BiLSTM module captures long-range sequential dependencies within SMILES strings, modeling how the ordering of chemical units affects global properties. Removing the LSTM component degraded performance across all properties, indicating that sequential context contributes to understanding the overall chemical environment of repeating units.

The parallel integration of CNN and BiLSTM thus provides complementary representations of local motifs and global sequence—a combination that is difficult to achieve with either architecture alone.

3.2 Importance of Explicit Topological Descriptors

While SMILES provides an efficient 1D representation, it does not explicitly encode three-dimensional topological information such as side-chain length, branching density, or spatial bulkiness. According to polymer physics, properties such as R_g and density are fundamentally controlled by chain topology and steric effects of side chains.

As shown in the ablation analysis (Section 2.2.2), the pure sequence-based variant ("Only SMILES Sequence") yielded the poorest performance for R_g and density. Incorporating domain-aware descriptors extracted via RDKit—including the ratio of side-chain atoms to backbone atoms and rotatable bond counts—reduced the R_g MAE from 1.535 Å to 1.322 Å (13.9% improvement). This indicates

that explicitly calculated topological descriptors provide geometric constraints that sequence encoders alone struggle to learn implicitly.

3.3 Contribution of Pretrained Embeddings and Data Augmentation

Data scarcity is a persistent challenge in polymer informatics, particularly for properties like T_c that require expensive simulations or measurements. To partially address this, we incorporated MolFormer embeddings pretrained on approximately 1.1 billion small molecules. The ablation results show modest but consistent improvements from MolFormer across most properties, with the largest effect on T_c (11.7% MAE reduction). This suggests that general chemical knowledge transfers partially to polymer property prediction, though the domain gap between small molecules and polymers likely limits the benefit.

Data augmentation through non-canonical SMILES enumeration proved important for generalization. By exposing the model to multiple valid SMILES representations of the same molecule during training, the network learns permutation-invariant molecular representations rather than memorizing specific character sequences. Removing augmentation increased T_c MAE by 19.7% and FFV MAE by 10.4%, confirming its role in preventing overfitting, particularly for properties with limited training data.

3.4 Limitations

Several limitations of the current work should be acknowledged.

Dataset scope. Our evaluation is confined to the NeurIPS 2025 challenge dataset. Generalization to other polymer datasets with different chemical spaces or property ranges remains to be validated. Direct comparison with prior methods is hindered by the lack of standardized benchmarks in polymer informatics; future work will evaluate Poly-Prophet on established datasets such as PI1M to enable more rigorous comparison.

Representation constraints. The model operates on monomer SMILES strings representing idealized repeating units. Real-world bulk polymers exhibit conformational heterogeneity, intermolecular entanglements, and molecular weight distributions that are not captured by monomeric representations. Properties sensitive to these factors may be systematically mispredicted.

Absence of 3D information. SMILES representations do not encode three-dimensional conformational information. While the backbone/sidechain descriptors partially address topological features, properties influenced by specific 3D conformations (e.g., crystallinity, mechanical properties) cannot be accurately predicted from 1D representations alone.

Domain mismatch in pretrained embeddings. MolFormer was pretrained on small molecules rather than polymers. While the ablation studies indicate positive transfer, the magnitude of improvement is modest, suggesting that polymer-specific pretraining could yield greater benefits.

Single-model architecture. Our competition ranking (top 23%) was achieved with a standalone model. Top-ranking solutions typically employ ensembles of diverse architectures; our single-model approach prioritizes interpretability over maximum predictive performance.

3.5 Future Directions

Several extensions could address the limitations identified above.

3D structural information. Incorporating geometric features from 3D molecular graphs or brief molecular dynamics simulations could improve prediction of conformation-dependent properties. Graph neural networks operating on 3D atomic coordinates represent a promising direction.

Molecular weight dependence. Feeding molecular weight parameters (e.g., number-average or weight-average molecular weight) as explicit inputs could improve prediction of properties sensitive to chain length, such as mechanical strength or viscosity.

Polymer-specific pretraining. Developing language models pretrained specifically on polymer SMILES, rather than small molecules, could provide more relevant chemical priors and reduce domain mismatch.

Inverse design. The current framework focuses on forward prediction (structure to property). Integrating the learned representations with generative models could enable inverse design—identifying

polymer structures that satisfy target property specifications. This remains an important goal for accelerating materials discovery.

4 Methodology

We present a hybrid deep learning framework for predicting five macroscopic polymer properties directly from repeat-unit SMILES strings. The data processing pipeline and model architecture are described in the following sections.

4.1 Dataset and Target Properties

The primary dataset is derived from the NeurIPS 2025 Open Polymer Prediction Challenge.[29] It comprises polymer repeat-unit SMILES strings paired with five target properties obtained via molecular dynamics (MD) simulations: glass transition temperature (T_g), thermal conductivity (T_c), fractional free volume (FFV), density (ρ), and radius of gyration (R_g).

To expand the training corpus and improve model generalization, three supplementary datasets provided by the competition organizers (targeting T_c, T_g, and FFV) were integrated. During data fusion, duplicate SMILES entries within each supplementary subset were averaged to yield a single consensus value. These subsets were then merged with the primary dataset: missing property values for overlapping SMILES were filled without overwriting existing data, while structurally novel SMILES were appended as independent training samples.

4.2 Data Preprocessing

We implemented a multi-stage preprocessing pipeline to ensure structural validity, data consistency, and stable gradient propagation across all target properties.

4.2.1 SMILES Cleaning and Standardization

Raw SMILES strings were systematically curated. Samples containing ambiguous R-group placeholders (e.g., [R], [R1]–[R5]) were discarded, as they denote incomplete structural definitions. SMILES strings that could not be parsed by the RDKit chemoinformatics library were excluded. All remaining valid structures were canonicalized using RDKit to ensure a unique, permutation-invariant string representation for each molecule.

4.2.2 Target Variable Scaling

A practical challenge in multi-target regression is the disparate magnitude of target variables. Specifically, FFV and T_c exhibit smaller numerical ranges compared to T_g. To prevent gradient imbalance and ensure comparable loss contributions during training, FFV values were scaled by a factor of 1000 and T_c values by 100. The remaining targets (T_g, ρ, R_g) were kept unscaled. These scaling transformations were reversed during inference to recover predictions in original physical units.

4.3 Feature Engineering

To avoid the limitations of relying on a single molecular representation, our framework extracts three complementary feature sets that encode chemical information at distinct levels: (i) character-level sequential features derived from SMILES tokenization; (ii) domain-aware physicochemical descriptors that distinguish the polymer backbone from side chains; and (iii) pretrained molecular embeddings extracted from MolFormer-XL. These multi-modal feature streams are concatenated and fed into the prediction layers (described in Section 4.4). The contribution of each feature source is quantified via ablation studies in Section 2.2.2.

4.3.1 SMILES Tokenization Features

SMILES strings encode polymer topologies as sequential text. We tokenized each repeat-unit SMILES at the character level using a fixed vocabulary of 45 discrete tokens. While this character-level parsing

splits multi-character atom symbols (e.g., "Cl" is processed as separate "C" and "l" tokens), the downstream sequence encoders have sufficient capacity to resolve these structural semantics implicitly. This strategy avoids the need for complex, domain-specific tokenizers. Each token is mapped to a 256-dimensional continuous vector via a learnable embedding layer, followed by dropout ($p = 0.025$) for regularization.

4.3.2 Backbone–Sidechain Structural Descriptors

Unlike small molecules, the macroscopic properties of polymers are strongly influenced by the structural distinction between the main backbone and pendant side chains. To capture this contrast explicitly, we extracted structural descriptors for each region of the repeat unit. The polymer backbone was identified using RDKit by locating the two wildcard atoms (*) within the SMILES and computing the shortest path connecting their adjacent heavy atoms. All remaining heavy atoms were classified as side-chain components.

For all prediction tasks, we computed a base set of six descriptors: atomic count and molecular weight for both backbone and side chain, plus the side-chain-to-backbone ratio of both metrics. These characterize the overall mass and size distribution within the repeat unit. Beyond this base set, we included target-specific descriptors informed by the underlying physics of each property:

- **FFV**: Rotatable bond counts for both backbone and side chains, as chain flexibility affects packing efficiency and free volume.
- R_g: Rotatable bond counts and saturated ring counts for both regions, since chain flexibility and cyclic rigidity jointly influence molecular conformation.
- ρ: Hydrogen bond donor and acceptor counts, as hydrogen bonding promotes dense molecular packing.
- T_g **and** T_c: Restricted to the base descriptors, relying on the sequence encoders to capture the complex rigidity patterns governing thermal properties.

This target-aware feature selection avoids introducing irrelevant descriptors and ensures that geometric inputs align with the physical nature of each target.

4.3.3 Pretrained Molecular Representations (MolFormer-XL)

To complement the handcrafted structural descriptors, we integrated pretrained representations from MolFormer-XL,[27] a Transformer model pretrained on approximately 1.1 billion SMILES strings. These embeddings encode general chemical semantics that may capture molecular patterns not explicitly represented by RDKit descriptors.

For each polymer SMILES, we extracted hidden states from the final layer of MolFormer-XL. The CLS token representation and the mean-pooled token representations were averaged to yield a 768-dimensional global embedding. To reduce the risk of overfitting and limit parameter overhead, this vector was truncated to its first 64 dimensions. The resulting vector was processed through a fully connected layer (projecting from 64 to 32 dimensions) with ReLU activation and dropout before fusion with the sequential and descriptor feature streams.

We note that MolFormer-XL was pretrained primarily on small molecules rather than polymers. While this introduces potential domain mismatch, the ablation studies in Section 2.2.2 indicate that the pretrained embeddings provide modest but consistent improvements, suggesting that general chemical priors transfer partially to polymer property prediction.

4.4 Model Architecture and Fusion Strategy

To integrate the multi-modal features described above, we designed a parallel neural network architecture termed Poly-Prophet. The network comprises a dual-branch sequence encoder, auxiliary feature projectors, and a multi-layer perceptron (MLP) fusion module.

4.4.1 Dual-Branch Sequence Encoder

The tokenized SMILES sequence is first mapped into a continuous space via a 256-dimensional learnable embedding layer with dropout regularization ($p = 0.025$). The embedded sequence is then processed by two parallel branches:

1. Multi-Scale 1D-CNN Branch: To capture local chemical motifs of varying sizes, this branch employs six parallel 1D convolutional layers with diverse kernel sizes ($k \in \{3, 5, 7, 15\}$) and dilations ($d \in \{1, 2\}$), creating multi-scale receptive fields. This design enables recognition of both compact functional groups and extended topological patterns. Following ReLU activation, each convolutional output undergoes both global max pooling (GMP) and global average pooling (GAP). GMP detects the presence of specific motifs, while GAP quantifies their density across the sequence. The pooled representations are concatenated to form a 704-dimensional local feature vector.

2. BiLSTM with Attention Branch: To model long-range dependencies and overall chain context, the embeddings are fed into a 2-layer bidirectional LSTM (BiLSTM) with hidden size 128 per direction. The 256-dimensional BiLSTM output sequence is processed by an additive self-attention mechanism to focus on structurally influential regions. A padding mask is applied prior to softmax to exclude padded tokens from attention computation. The context-weighted sum yields a 256-dimensional global sequential vector.

4.4.2 Auxiliary Feature Processors

Directly concatenating disparate feature modalities can cause optimization imbalance due to differing dimensionalities and statistical distributions. To address this, continuous auxiliary features are processed through dedicated projection modules. The backbone/sidechain descriptors and the 64-dimensional MolFormer embeddings are independently passed through nonlinear projection networks (Dense $\rightarrow$ ReLU $\rightarrow$ Dropout), down-projecting to 64 and 32 dimensions respectively. This ensures that auxiliary features provide concentrated structural information without dominating the higher-dimensional sequence features during optimization.

4.4.3 Multi-Modal Fusion and Prediction Head

The outputs from all four processing streams—CNN features (704-dim), attention-weighted BiLSTM features (256-dim), projected RDKit descriptors (64-dim), and projected MolFormer features (32-dim)—are concatenated to form a 1056-dimensional multi-modal representation.

This fused vector is fed into a cascaded MLP prediction head with progressively decreasing layer sizes ([512, 256, 128]). Each dense layer is followed by layer normalization, ReLU activation, and dropout regularization. A final linear layer projects the 128-dimensional hidden state to a single continuous output representing the predicted property. Mean squared error (MSE) was used as the training objective, optimized with AdamW and a ReduceLROnPlateau learning rate scheduler.

Declarations

Data Availability

The dataset analysed in this study is from the NeurIPS 2025 Open Polymer Prediction Challenge and is publicly available at https://kaggle.com/competitions/neurips-open-polymer-prediction-2025.

Code Availability

The source code and trained model weights are available at https://www.kaggle.com/code/bfladderbean/neurips-open-polymer-prediction.

Author Contributions

S.L., J.J.: Conceptualization, Supervision, Formal analysis, Data curation, Methodology. S.F.: Data Curation, Investigation, Validation, Visualization. S.L., J.J., S.F.: Writing – original draft, Writing – review & editing.

Competing Interests

The authors declare no competing interests.

Use of AI tools in manuscript preparation

During the preparation of this work, the authors used the large language model Gemini 3.1 Pro solely for English-language polishing and readability. The tool was not used to generate data, draw scientific

conclusions, or create figures. The author reviewed and edited all content and takes full responsibility for the integrity of the publication.

Funding

All costs of this scientific report, including publication fees, were borne personally by the corresponding authors. No external financial support was obtained.

References

[1] F. Zhao, Y. Shi, L. Pan, and G. Yu, "Multifunctional nanostructured conductive polymer gels: Synthesis, properties, and applications," *Accounts of Chemical Research*, vol. 50, pp. 1734–1743, July 2017.

[2] H. Tran, R. Gurnani, C. Kim, G. Pilania, H.-K. Kwon, R. P. Lively, and R. Ramprasad, "Design of functional and sustainable polymers assisted by artificial intelligence," *Nature Reviews Materials*, vol. 9, pp. 866–886, Dec. 2024.

[3] C. Kim, R. Batra, L. Chen, H. Tran, and R. Ramprasad, "Polymer design using genetic algorithm and machine learning," *Computational Materials Science*, vol. 186, p. 110067, Jan. 2021.

[4] T. D. Huan, A. Mannodi-Kanakkithodi, C. Kim, V. Sharma, G. Pilania, and R. Ramprasad, "A polymer dataset for accelerated property prediction and design," *Scientific Data*, vol. 3, p. 160012, Mar. 2016.

[5] T. B. Martin and D. J. Audus, "Emerging trends in machine learning: A polymer perspective," *ACS Polym. Au*, vol. 3, pp. 239–258, June 2023.

[6] W. Ge, R. De Silva, Y. Fan, S. A. Sisson, and M. H. Stenzel, "Machine learning in polymer research," *Advanced Materials*, vol. 37, p. 2413695, Mar. 2025.

[7] M. Reis, F. Gusev, N. G. Taylor, S. H. Chung, M. D. Verber, Y. Z. Lee, O. Isayev, and F. A. Leibfarth, "Machine-learning-guided discovery of 19f MRI agents enabled by automated copolymer synthesis," *J. Am. Chem. Soc.*, vol. 143, pp. 17677–17689, Oct. 2021.

[8] M. A. Webb, N. E. Jackson, P. S. Gil, and J. J. de Pablo, "Targeted sequence design within the coarse-grained polymer genome," *Science Advances*, vol. 6, p. eabc6216, Oct. 2020.

[9] C. Kuenneth, J. Lalonde, B. L. Marrone, C. N. Iverson, R. Ramprasad, and G. Pilania, "Bioplastic design using multitask deep neural networks," *Communications Materials*, vol. 3, p. 96, Dec. 2022.

[10] H. Yamada, C. Liu, S. Wu, Y. Koyama, S. Ju, J. Shiomi, J. Morikawa, and R. Yoshida, "Predicting materials properties with little data using shotgun transfer learning," *ACS Cent. Sci.*, vol. 5, pp. 1717–1730, Oct. 2019.

[11] D. Weininger, "SMILES, a chemical language and information system. 1. introduction to methodology and encoding rules," *J. Chem. Inf. Comput. Sci.*, vol. 28, pp. 31–36, Feb. 1988.

[12] D. Weininger, A. Weininger, and J. L. Weininger, "SMILES. 2. algorithm for generation of unique SMILES notation," *J. Chem. Inf. Comput. Sci.*, vol. 29, pp. 97–101, May 1989.

[13] D. Weininger, "SMILES. 3. DEPICT. graphical depiction of chemical structures," *J. Chem. Inf. Comput. Sci.*, vol. 30, pp. 237–243, Aug. 1990.

[14] M. A. Parvez and I. M. Mehedi, "High-accuracy polymer property detection via pareto-optimized SMILES-based deep learning," *Polymers*, vol. 17, no. 13, 2025.

[15] R. Ma and T. Luo, "PI1m: A benchmark database for polymer informatics," *J. Chem. Inf. Model.*, vol. 60, pp. 4684–4690, Oct. 2020.

[16] H. Qiu and Z.-Y. Sun, "Exploring SMILES and BigSMILES: Toward efficient and scalable polymer representations for machine learning," *Macromolecules*, vol. 58, pp. 7084–7093, July 2025.

[17] A. A. Toropov and K. Roy, "QSPR modeling of lipid-water partition coefficient by optimization of correlation weights of local graph invariants," *J. Chem. Inf. Comput. Sci.*, vol. 44, pp. 179–186, Jan. 2004.

[18] A. A. Toropov and E. Benfenati, "SMILES in QSPR/QSAR modeling: Results and perspectives," *Current Drug Discovery Technologies*, vol. 4, no. 2, pp. 77–116, 2007.

[19] S. Dara and P. Tumma, "Feature extraction by using deep learning: A survey," in *2018 Second International Conference on Electronics, Communication and Aerospace Technology (ICECA)*, pp. 1795–1801, 2018.

[20] L. A. Miccio and G. A. Schwartz, "From chemical structure to quantitative polymer properties prediction through convolutional neural networks," *Polymer*, vol. 193, p. 122341, 2020.

[21] G. Chen, L. Tao, and Y. Li, "Predicting polymers' glass transition temperature by a chemical language processing model," *Polymers*, vol. 13, no. 11, 2021.

[22] C. Gao, X. Min, M. Fang, T. Tao, X. Zheng, Y. Liu, X. Wu, and Z. Huang, "Innovative materials science via machine learning," *Advanced Functional Materials*, vol. 32, no. 1, p. 2108044, 2022.

[23] X. Zhang, Y. Sheng, X. Liu, J. Yang, W. A. Goddard III, C. Ye, and W. Zhang, "Polymer-unit graph: Advancing interpretability in graph neural network machine learning for organic polymer semiconductor materials," *J. Chem. Theory Comput.*, vol. 20, pp. 2908–2920, Apr. 2024.

[24] J. Tamnanloo, A. H. Mokarizadeh, F. Toiserkani, H. Abeysinghe, A. Joy, and M. Tsige, "Graph neural networks for surface tension prediction of polymers: A comparative analysis with descriptor-based models," *Macromolecules*, vol. 58, pp. 3764–3773, Apr. 2025.

[25] S. Han, Y. Kang, H. Park, J. Yi, G. Park, and J. Kim, "Multimodal transformer for property prediction in polymers," *ACS Appl. Mater. Interfaces*, vol. 16, pp. 16853–16860, Apr. 2024.

[26] N. Aleb and N. Y. Abu-Thabit, "TransTg: a new transformer model for predicting glass transition temperature of polymers from monomers' molecular structures," *Neural Computing and Applications*, vol. 37, pp. 2733–2746, Feb. 2025.

[27] J. Ross, B. Belgodere, V. Chenthamarakshan, I. Padhi, Y. Mroueh, and P. Das, "Large-scale chemical language representations capture molecular structure and properties," *Nature Machine Intelligence*, vol. 4, pp. 1256–1264, 2022.

[28] Q. Li, S. Zhan, Z. Liu, C. Dong, H. Zhao, T. Yue, Q. Zhao, L. Zhang, Y. Li, and J. Liu, "Developing hybrid machine learning frameworks for polymer property prediction based on composition and sequence features," *J. Chem. Inf. Model.*, vol. 65, pp. 7478–7492, July 2025.

[29] G. Liu, J. Xu, E. Inae, Y. Zhu, Y. Li, T. Luo, M. Jiang, Y. Yan, W. Reade, S. Dane, A. Howard, and M. Cruz, "Neurips - open polymer prediction 2025." https://kaggle.com/competitions/neurips-open-polymer-prediction-2025, 2025. Kaggle.